

CONSOLIDATED TECHNICAL SPECIFICATION

Winding Machines

Soft Winding · Hard Winding · Sewing Thread

Machine configuration, yarn compatibility, bobbin geometry and finished-package requirements for three winding machine types, consolidated from the two issued specification documents.

MACHINE TYPES	TOTAL SPINDLES	MAX WINDING SPEED	SUPPLY FREQUENCY
3	156	1 000 m/min	50 Hz

01 Soft Winding 48 spindles Cotton · spun & filament polyester Bobbin 160.40 mm · cylindrical Package Ø 160 mm · 1 kg 0.30–0.40 g/cm ³ · open build	02 Hard Winding 60 spindles Cotton Bobbin 173.00 mm · conical Cone 150/180 mm · h 155 mm · 1 kg 0.40–0.50 g/cm ³ · dense build	03 Sewing Thread 48 spindles Spun polyester (2- and 3-ply) Bobbin 114.00 mm · conical 4 000 m / 100 g at 50/2 Tension control + auto doffing
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Compiled from	Machine Specification.pdf + Winding_Machine_Technical_Specifications.pdf
Drawings	Bobbin dimension drawings and the finished-package photograph are reproduced from the source documents
Units	All linear dimensions in millimetres (mm) unless stated otherwise

1. Scope & Machine Overview

What this document covers

This specification consolidates the requirements for three distinct winding machines. Although all three convert yarn from a supply form onto a defined carrier, each serves a different downstream purpose and is therefore specified independently: the **soft winding machine** builds open, permeable packages intended for package dyeing; the **hard winding machine** builds dense, firm cones for storage, transport and onward processing; and the **sewing thread machine** builds precision, length-controlled cones of finished sewing thread.

Sections 4 to 6 give the full per-machine data set, each with its dimensioned bobbin drawing. Section 7 restates the requirements for each machine type in narrative form, with the engineering intent behind each figure made explicit.

Contents

SECTION	TITLE	CONTENT
2	Common Electrical Specification	Supply voltage, phases, protective conductors, frequency
3	Comparison Matrix	All three machines compared parameter by parameter
4	Machine 01 — Soft Winding	48 spindles · yarn counts · bobbin 160.40 mm · soft package targets
5	Machine 02 — Hard Winding	60 spindles · yarn counts · bobbin 173.00 mm · hard cone targets
6	Machine 03 — Sewing Thread	48 spindles · yarn counts · bobbin 114.00 mm · thread length targets
7	Detailed Requirements by Machine Type	Purpose, build, process, control and acceptance criteria per machine
8	Notes on the Source Data	Transcription correction and items requiring confirmation

2. Common Electrical Specification

Applies identically to all three machine types

All three machines share a single electrical supply specification. The machine shall be suitable for either a three-phase or a single-phase supply as installed, in both cases with protective earth and neutral conductors provided.

SUPPLY ARRANGEMENT	VOLTAGE	CONDUCTORS	FREQUENCY
Three phase (3PH)	380 – 440 V AC	L1, L2, L3 + PE + N	50 Hz
Single phase (1PH)	220 – 230 V AC	L + PE + N	50 Hz

Installation note

The supply arrangement (3PH or 1PH) must be confirmed per installation site before manufacture, as it affects drive selection, cabling and the protective device rating. Both options require PE and N to be brought to the machine terminal box.

3. Comparison Matrix

All three machines, parameter by parameter

PARAMETER	SOFT WINDING	HARD WINDING	SEWING THREAD
Primary application	Package dyeing (intermediate package)	Storage, transport and onward processing	Finished sewing thread for direct use
Number of spindles	48	60	48
Yarn compatibility	Cotton; spun polyester; filament polyester	Cotton	Spun polyester
Cotton count range	10/1 – 50/1 20/2 – 100/2	10/1 – 50/1 20/2 – 100/2	—
Polyester count range	20/2 – 60/2 SSP (spun) 50 D – 300 D (filament)	—	20/2 – 100/2 SSP 16/3 – 80/3 SSP
Carrier form	Cylindrical, stepped ends	Conical (tapered)	Conical (tapered)
Carrier length	160.40 mm	173.00 mm	114.00 mm
Carrier end diameters	54.70 / 49.60 / 43.15 mm 43.15 / 48.80 / 52.90 mm	71.00 / 67.00 mm (large) 37.00 / 26.00 mm (small)	38.00 / 35.00 mm (large) 28.00 / 12.60 mm (small)
Finished package size	Ø 160 mm	Ø 150 mm top / Ø 180 mm base height 155 mm	Not specified
Finished package weight	1 kg	1 kg	Not specified
Target density	0.30 – 0.35 g/cm ³ (cotton) 0.35 – 0.40 g/cm ³ (polyester)	0.40 – 0.50 g/cm ³	Not specified
Length / weight target	—	—	4 000 m per 100 g (50/2 spun)
Maximum winding speed	1 000 m/min	1 000 m/min	1 400 rpm
Yarn tension control	Not specified	Not specified	Required
Automatic doffing / feeding	Not specified	Not specified	Required
Rated voltage	3PH 380–440 V AC or 1PH 220–230 V AC	3PH 380–440 V AC or 1PH 220–230 V AC	3PH 380–440 V AC or 1PH 220–230 V AC
Supply frequency	50 Hz	50 Hz	50 Hz

“Not specified” indicates the parameter was not stated in either source document for that machine; it is listed so that the gap is visible rather than assumed. “—” indicates the parameter does not apply to that machine. Carrier end diameters are given as stepped / stacked dimensions exactly as drawn.

MACHINE 01

Soft Winding Machine

48 spindles · open, permeable packages for package dyeing

SPINDLES	PACKAGE Ø	PACKAGE WEIGHT	MAX SPEED
48	160 mm	1 kg	1 000 m/min

4.1 Machine and electrical data

SPINDLES	FREQUENCY	RATED VOLTAGE
48	50 Hz	3PH (380–440 V AC) or 1PH (220–230 V AC) with PE & N

4.2 Yarn compatibility and count range

YARN TYPE	COUNT RANGE
Cotton yarn	10/1 to 50/1 & 20/2 to 100/2
Spun polyester	20/2 to 60/2 SSP
Filament polyester	50 D to 300 D

4.3 Finished package requirements

PARAMETER	REQUIREMENT
Finished package diameter	160 mm
Finished package weight	1 kg
Target package density — cotton	0.30 – 0.35 g/cm ³
Target package density — polyester	0.35 – 0.40 g/cm ³
Winding speed	1 000 m/min (maximum)

4.4 Bobbin specification

OVERALL LENGTH	MAIN BODY LENGTH	END SECTIONS
160.40 mm	143.00 mm	10.40 mm / 7.00 mm
LEFT-END DIAMETERS	RIGHT-END DIAMETERS	CARRIER TYPE
54.70 / 49.60 / 43.15 mm	43.15 / 48.80 / 52.90 mm	Soft winding bobbin

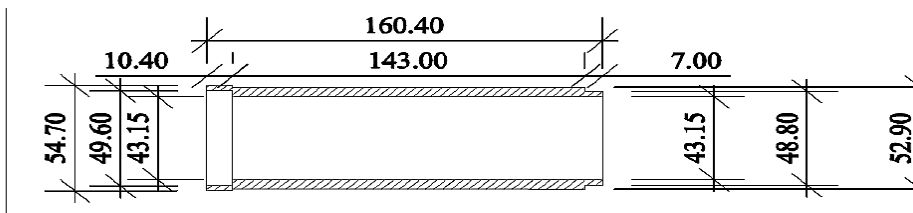


Figure 1 — Soft winding bobbin, dimensioned drawing (all dimensions in mm). Reproduced from the source specification.

MACHINE 02

Hard Winding Machine

60 spindles · dense, firm cones for storage and onward processing

SPINDLES	CONE HEIGHT	PACKAGE WEIGHT	MAX SPEED
60	155 mm	1 kg	1 000 m/min

5.1 Machine and electrical data

SPINDLES	FREQUENCY	RATED VOLTAGE
60	50 Hz	3PH (380–440 V AC) or 1PH (220–230 V AC) with PE & N

5.2 Yarn compatibility and count range

YARN TYPE	COUNT RANGE
Cotton yarn	10/1 to 50/1 & 20/2 to 100/2

This machine is specified for cotton only. Polyester — spun or filament — is not within its stated compatibility range.

5.3 Finished package requirements

PARAMETER	REQUIREMENT
Package diameter — top	150 mm
Package diameter — bottom	180 mm
Package height	155 mm
Package weight	1 kg
Target hardness / density	0.40 – 0.50 g/cm ³
Winding speed	1 000 m/min (max.)



Figure 2 — Hard-wound finished package: 150 mm top Ø, 180 mm base Ø, 155 mm height.

5.4 Bobbin specification

OVERALL LENGTH	LARGE-END DIAMETERS	SMALL-END DIAMETERS
173.00 mm	71.00 / 67.00 mm	37.00 / 26.00 mm

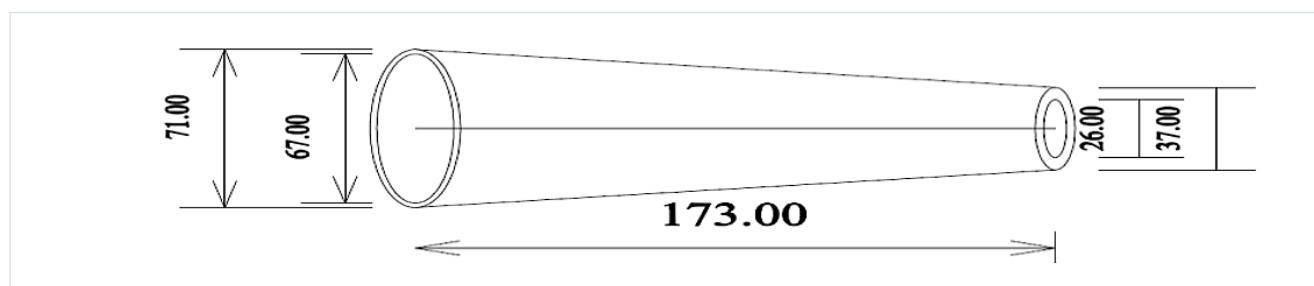


Figure 3 — Hard winding bobbin, dimensioned drawing (all dimensions in mm). Reproduced from the source specification.

MACHINE 03

Sewing Thread Machine

48 spindles · precision, length-controlled finished sewing thread

SPINDLES	LENGTH / WEIGHT	MAX SPEED	AUTO DOFFING
48	4 000 m / 100 g	1 400 rpm	Required

6.1 Machine and electrical data

SPINDLES	FREQUENCY	RATED VOLTAGE
48	50 Hz	3PH (380–440 V AC) or 1PH (220–230 V AC) with PE & N

6.2 Yarn compatibility and count range

YARN TYPE	COUNT RANGE
Spun polyester	20/2 to 100/2 SSP & 16/3 to 80/3 SSP

6.3 Thread and process requirements

PARAMETER	REQUIREMENT
Thread length / weight	4 000 m per 100 g (yarn count 50/2 spun)
Required winding speed	1 400 rpm (maximum)
Yarn tension control	Required
Automatic doffing / feeding	Required

6.4 Bobbin specification

OVERALL LENGTH	LARGE-END DIAMETERS	SMALL-END DIAMETERS
114.00 mm	38.00 / 35.00 mm	28.00 / 12.60 mm

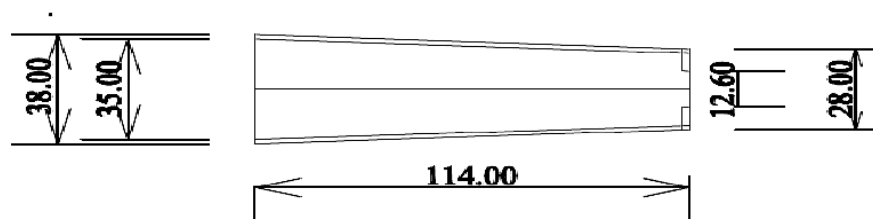


Figure 4 — Sewing thread bobbin, dimensioned drawing (all dimensions in mm). Reproduced from the source specification.

7. Detailed Requirements by Machine Type

Purpose, build, process behaviour and acceptance criteria

The preceding sections present the specification as tabulated data. This section states the same requirements in narrative form, with the engineering intent behind each figure made explicit. It introduces no new numeric requirements.

7.1 Soft Winding Machine

Purpose and process role

The soft winding machine produces packages whose defining property is **permeability** rather than compactness. The package is an intermediate product: it is built specifically so that dye liquor can be forced through the full wound mass during package dyeing. Every requirement below follows from that single objective — the yarn must be laid down firmly enough to hold its shape through handling and loading into a dyeing carrier, yet loosely enough that liquor reaches the innermost layers at the same rate as the outermost ones.

Capacity and configuration

- The machine shall be built with **48 winding spindles**, each independently threaded and each capable of being stopped without interrupting the remaining spindles.
- Electrical supply shall be **3PH 380–440 V AC or 1PH 220–230 V AC at 50 Hz**, with protective earth and neutral provided.

Yarn compatibility

This is the most versatile of the three machines and shall handle three distinct fibre classes without mechanical reconfiguration beyond normal setting changes: **cotton** in counts 10/1 to 50/1 and 20/2 to 100/2; **spun polyester** in counts 20/2 to 60/2 SSP; and **filament polyester** from 50 D to 300 D. The span from a coarse 10/1 cotton to a fine 300 D filament is wide, so yarn-path components — guides, tensioners and traverse elements — shall be selected to run the full declared range without substitution.

Package build and density

The finished package shall be **160 mm in diameter** and weigh **1 kg**. Target density is fibre-dependent and is the primary quality acceptance criterion:

- **Cotton: 0.30 – 0.35 g/cm³**. Cotton swells appreciably when wet, so the dry wound density is held lower to leave room for that swelling without choking liquor flow.
- **Polyester: 0.35 – 0.40 g/cm³**. Polyester does not swell comparably, so a slightly firmer build is permitted — and is desirable for package stability.

Density shall be uniform through the package. A package that meets the target as an average while varying materially between core and surface will dye unevenly and does not satisfy this specification.

Speed and carrier

Maximum winding speed shall be **1 000 m/min**. Yarn shall be wound onto the cylindrical stepped bobbin detailed in section 4.4: **160.40 mm** overall, with a **143.00 mm** main body and stepped end sections of 10.40 mm and 7.00 mm. The stepped ends (54.70 / 49.60 / 43.15 mm at one end, 43.15 / 48.80 / 52.90 mm at the other) allow bobbins to nest and to locate positively on the dyeing carrier spindle.

7.2 Hard Winding Machine

Purpose and process role

The hard winding machine has the opposite objective to the soft winder. Here the package is a **finished or near-finished** product, built for maximum yarn mass in minimum volume and for mechanical stability in storage, handling and transport. Liquor permeability is irrelevant; what matters is a firm, stable cone that will not collapse, telescope or shed layers when stacked, and that unwinds cleanly downstream.

Capacity and configuration

- The machine shall be built with **60 winding spindles** — the highest spindle count of the three machines.
- Electrical supply shall be **3PH 380–440 V AC or 1PH 220–230 V AC at 50 Hz**, with protective earth and neutral provided.

Yarn compatibility

The machine is specified for **cotton only**, in counts 10/1 to 50/1 and 20/2 to 100/2. Unlike the soft winder, no polyester capability is required. This narrower remit permits the yarn path and tension system to be optimised for the higher tension that hard winding demands.

Package build and hardness

The finished package is a **tapered cone**, not a cylinder, and shall be built to the following envelope: **150 mm diameter at the top, 180 mm diameter at the base and 155 mm overall height**, at a finished weight of **1 kg**. Target hardness is **0.40 – 0.50 g/cm³** — between roughly 15 % and 65 % denser than the soft-wound equivalent. Achieving this reliably requires correspondingly higher and more tightly regulated winding tension, and traverse control good enough to hold square, stable cone edges at that density; soft or broken-down edges are the usual failure mode of an under-controlled hard winder.

Speed and carrier

Maximum winding speed shall be **1 000 m/min**, matching the soft winder. Yarn shall be wound onto the conical bobbin detailed in section 5.4: **173.00 mm overall** — the longest carrier of the three — tapering from a large end of 71.00 / 67.00 mm to a small end of 37.00 / 26.00 mm. That taper is what generates the finished cone profile, so carrier geometry and the 150 / 180 / 155 mm package envelope must be verified together rather than separately.

7.3 Sewing Thread Machine

Purpose and process role

The sewing thread machine winds **finished sewing thread** for sale and direct use in sewing operations. Its governing requirement is neither permeability nor density but **precision and repeatability**. A sewing thread package is consumed at high speed by a machine that cannot tolerate tension spikes, snarls or sloughing, and it is sold on a declared length. Both constraints drive the specification below.

Capacity and configuration

- The machine shall be built with **48 winding spindles**.
- Electrical supply shall be **3PH 380–440 V AC or 1PH 220–230 V AC at 50 Hz**, with protective earth and neutral provided.

Yarn compatibility

The machine is specified for **spun polyester** only, in counts 20/2 to 100/2 SSP and 16/3 to 80/3 SSP. The inclusion of three-ply constructions (16/3 to 80/3) is significant: three-ply sewing threads are torque-prone, and the yarn path shall be arranged to wind them without introducing or releasing twist that would cause snarling at the point of use.

Length and weight control

The thread length-to-weight requirement is **4 000 m per 100 g** at a reference count of **50/2 spun** — equivalently 40 m per gram. This is the commercial basis on which the package is sold, so the machine shall provide length measurement and cut-off accurate enough to guarantee the declared length on every package, not merely on average across a batch.

Tension control

Active yarn tension control is a mandatory requirement for this machine. Consistent tension governs both the length-per-weight accuracy above and the thread's residual elongation and strength, which must not vary within a package or between packages. Open-loop or purely mechanical tensioning that drifts as the package builds does not meet this requirement.

Automation

Automatic doffing and automatic feeding are mandatory requirements. The machine shall eject completed packages and introduce empty carriers without manual intervention, and shall take up supply yarn automatically. Across 48 spindles this is what makes continuous unattended running practical; it also removes the package-to-package variation that manual doffing introduces at the start and end of each build.

Speed and carrier

Required winding speed is **1 400 rpm maximum**. Note that this figure is stated as a rotational speed, whereas the soft and hard winders are specified in linear terms (1 000 m/min) — see section 8.2. Yarn shall be wound onto the conical bobbin detailed in section 6.4: **114.00 mm** overall — the shortest carrier of the three — tapering from a large end of 38.00 / 35.00 mm to a small end of 28.00 / 12.60 mm.

8. Notes on the Source Data

Transcription correction and items requiring confirmation

The source data is reproduced as issued. Points identified during consolidation are recorded here rather than silently resolved.

8.1 Transcription correction applied

ITEM	AS ISSUED	AS PRESENTED HERE	BASIS
Hard winder target density	0.40 – .050 gm/cm ³	0.40 – 0.50 g/cm ³	“.050” is an evident typographical error; 0.050 g/cm ³ would be an order of magnitude <i>lower</i> than the soft-winding target and inconsistent with a hard, dense package

8.2 Items recommended for confirmation

- **Speed units are inconsistent between machines.** The soft and hard winders are specified at 1 000 m/min (linear yarn speed); the sewing thread machine at 1 400 rpm (rotational speed). These are not directly comparable, and the rotational figure cannot be converted to a linear speed without a stated reference diameter. Confirm whether 1 400 rpm is the intended basis for the sewing thread machine.
- **The sewing thread package envelope is not specified.** No finished package diameter, height or weight is given for the sewing thread machine, although both other machines have a defined envelope and a 1 kg finished weight.
- **Tension control and automatic doffing are stated only for the sewing thread machine.** Confirm whether these are genuinely not required on the soft and hard winders, or were simply not recorded for them.
- **Soft bobbin length and soft package diameter are both nominally 160 mm.** These are independent dimensions — 160.40 mm is axial length, 160 mm is wound diameter — and the coincidence is not a relationship.
- **Density basis is not stated.** Confirm whether the target densities are to be measured on the wound yarn mass alone or on the gross package volume including the carrier, as the two give materially different figures.

8.3 Source documents

SOURCE FILE	CONTRIBUTED CONTENT
Machine Specification.pdf	Finished-package requirements, target densities, winding speeds, thread length/weight, tension and auto-doffing. Package photograph (Figure 2).
Winding_Machine_Technical_Specifications.pdf	Machine configuration: spindle counts, electrical specification, yarn compatibility and count ranges, and all bobbin dimension drawings (Figures 1, 3 and 4).

End of document

All values in sections 1 to 6 are reproduced unaltered, except as recorded in section 8.1.